

Jacobian of Alpoge's candidate counterexample map $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$

$$F(x, y, z) = \left((1 + xy)^3 z + y^2(1 + xy)(4 + 3xy), \quad y + 3x(1 + xy)^2 z + 3xy^2(4 + 3xy), \quad 2x - 3x^2 y - x^3 z \right)$$

$$\frac{\partial F}{\partial(x, y, z)} = \begin{pmatrix} 3y^3(xy + 1) + y^3(3xy + 4) + 3yz(xy + 1)^2 & 3xy^2(xy + 1) + xy^2(3xy + 4) + 3xz(xy + 1)^2 + 2y(xy + 1)(3xy + 4) & (xy + 1)^3 \\ 9xy^3 + 6xyz(xy + 1) + 3y^2(3xy + 4) + 3z(xy + 1)^2 & 9x^2y^2 + 6x^2z(xy + 1) + 6xy(3xy + 4) + 1 & 3x(xy + 1)^2 \\ -3x^2z - 6xy + 2 & -3x^2 & -x^3 \end{pmatrix}$$

$$\det\left(\frac{\partial F}{\partial(x, y, z)}\right) = -2 \quad (\text{SymPy-verified identity in } \mathbb{Q}[x, y, z])$$

Three distinct points collapse to the same image:

$$F\left(0, 0, -\frac{1}{4}\right) = F\left(1, -\frac{3}{2}, \frac{13}{2}\right) = F\left(-1, \frac{3}{2}, \frac{13}{2}\right) = \left(-\frac{1}{4}, 0, 0\right)$$

Demo: $\det J$ evaluated at 54 small integer points from $[-5, 5]^3$ (seed 42), each returning -2 . This is illustrative only — the script's rigorous MC uses $[-1000, 1000]^3$, $k=30$, for a Schwartz–Zippel false-positive bound $\lesssim 2.2 \times 10^{-65}$.

$\det J = -2$ at each of the following (x, y, z) :

(5, -4, -5)	(-1, -2, -2)	(-3, -4, 5)	(3, -4, 4)	(1, -5, -5)	(-4, -2, -2)
(3, 4, -5)	(3, -2, 5)	(3, 1, -2)	(2, 4, -1)	(-5, -3, 1)	(0, -1, -3)
(-2, 0, -4)	(-4, 1, -4)	(0, 0, 4)	(-1, -5, 2)	(3, -4, 1)	(-4, 3, -1)
(5, 4, 0)	(4, -2, -4)	(-5, 5, -2)	(-1, -4, -2)	(-4, 1, -1)	(2, 5, 0)
(-3, 0, 0)	(-2, 5, -1)	(5, 5, -4)	(4, 5, -3)	(3, -2, -3)	(2, 1, -1)
(5, 3, -2)	(5, 0, -5)	(-2, -5, 0)	(1, -1, -4)	(-2, 4, 0)	(-2, 5, 2)
(1, 5, 2)	(-3, -1, -3)	(-2, 3, 3)	(-1, 4, 1)	(4, 1, 0)	(-2, -3, 3)
(2, -4, -5)	(-4, -3, 5)	(-3, 5, 1)	(4, -4, 1)	(1, 4, 2)	(3, -1, 3)
(-5, 5, -4)	(5, 3, -1)	(5, 0, -4)	(-1, 1, -3)	(2, -5, -1)	(3, -3, 3)